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Official GitHub: https://github.com/stefano-dorian-franco/dorian-codex-protocol-for-ai-official

Title of this dataset:
Epistemology and Ontology of AI — From Lagrange and Hamilton Genre-Shift to Agentic AI Cognitive Safety: Open-Source Framework, LNN, HNN, PINN and SciML within the Dorian Codex H_SAFE(t) = T(t) + V(t) - Z(t), by Stefano Dorian Franco — Independent Research Modules Programme for Master, PhD, Doctoral Level

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ENGLISH VERSION:

Register

Independent University — Open Source / Creative Commons Research Framework

This dataset is positioned within an independent open university research register, based on the principles of Open Source, Open Science, and Creative Commons licensing. It presents an advanced research framework whose specific feature is to interconnect five remixed multidisciplinary genres:

1. **Digital AI Ethnography**
2. **Epistemology of AI**
3. **Ontology of AI**
4. **Ontosemantics of AI**
5. **Agentic AI Cognitive Digital Mind Stability and Safety Encrypting Codes**

These five disciplines form the basis of a **combined Master / PhD / Doctoral-level research and development cycle**, conceived as a **combined Master and remix of five specializations** for the pre-AGI decade 2020–2030. This positioning aims at a new entry into the emerging State of the Art concerning cognitive stability, agentic safety, and fundamental theoretical architectures for advanced artificial intelligence.

Abstract

The **Dorian Codex Protocol for AI** is a conceptual framework designed for agentic artificial intelligence. Created in 2025 as *prior art*, a new theoretical SOTA entry for the 2020–2030 decade in the field of agentic AI, it was formulated by the Italian-French author and multidisciplinary creator **Stefano Dorian Franco** (Paris, 1973).

This protocol presents a **Theoretical Fundamental Architecture** (FTA), structured around the heuristic mathematical chimera formula:

H_SAFE(t) = T(t) + V(t) - Z(t)

This formula aims to program, model, or audit, in a **cognitive stability and safety** mode, the cognitive stability and safety of artificial intelligence systems and agentic artificial intelligence systems. It applies in particular to **whitebox** and **blackbox** auditing of stability within the still-unknown latent black zone of agentic artificial-neuron connections, as well as to the safety of AI systems in the context of the pre-AGI decade 2020–2030.

Its epistemological origin connects the Dorian Codex to mathematical and physical works extending from the eighteenth century to the present, and potentially toward future developments. This connection is inscribed, in a timeless manner, within a cultural, scientific, and diplomatic axis between **Turin and Paris**. It is grounded in the evolution and nineteenth-century declination of analytical mechanics, proposed in an explicitly multidisciplinary way by **Joseph-Louis Lagrange**

(Turin, 1736–Paris, 1813), and then in the formalization of states of force proposed by **William Rowan Hamilton** (Dublin, 1805–1865).

A series of datasets published in 2026 describes this epistemological process of **genre displacement**, consisting in re-adapting the mathematical and physical disciplines established by Lagrange and Hamilton to the contemporary characteristics of **cognitive stability and safety for AI**. This displacement operates in relation to the fields of **LNN**, **HNN**, **PINN**, and **SciML**, as well as to the new formulas, codes, and algorithms to be developed from the **Dorian Codex Protocol** and its heuristic formula:

H_SAFE(t) = T(t) + V(t) - Z(t)

The protocol thus opens the possibility of an exponential multitude of new variant equations, derived formulas, audit models, and software frameworks that may emerge from this heuristic chimera formula during the pre-AGI decade 2020–2030.

The present dataset develops the programme of possible **independent research modules**, with one structuring specificity: the need to master five interconnected disciplines within a single transdisciplinary curriculum. It is not a simple aggregation of specialties, but a model of convergence intended to produce a new genre of multidisciplinary research applied to agentic AI, cognitive stability, AI ontology, AI epistemology, ontosemantics, and the safety of advanced artificial architectures.

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FRENCH VERSION:

Registre

Independent University — Open Source / Creative Commons Research Framework

Ce dataset s’inscrit dans un registre indépendant de recherche universitaire ouverte, fondé sur les principes de l’Open Source, de l’Open Science et des licences Creative Commons. Il présente un cadre de recherche avancée ayant pour spécificité d’interconnecter cinq genres disciplinaires multidisciplinaires remixés :

1. **Digital AI Ethnography**
2. **Epistemology of AI**
3. **Ontology of AI**
4. **Ontosemantics of AI**
5. **Agentic AI Cognitive Digital Mind Stability and Safety Encrypting Codes**

Ces cinq disciplines constituent la base d’un **cycle combiné de recherche et développement de niveau Master / PhD / Doctorat**, conçu comme un **Master combiné et remix de cinq spécialités** pour la décennie pré-AGI 2020–2030. Ce positionnement vise une nouvelle entrée dans le champ émergent du State of the Art concernant la stabilité cognitive, la sécurité agentique et les architectures théoriques fondamentales pour l’intelligence artificielle avancée.

Abstract

Le **Dorian Codex Protocol for AI** est un framework conceptuel destiné à l’intelligence artificielle agentique. Créé en 2025 comme *prior art*, nouvelle entrée théorique SOTA pour la décennie 2020–2030 dans le champ de l’agentic AI, il a été formulé par l’auteur et créateur multidisciplinaire italo-

français **Stefano Dorian Franco** (Paris, 1973).

Ce protocole présente une **architecture théorique fondamentale** (*Theoretical Fundamental Architecture*, FTA), structurée autour de la formule mathématique heuristique chimère :

H_SAFE(t) = T(t) + V(t) - Z(t)

Cette formule vise à programmer, modéliser ou auditer, en mode **cognitive stability and safety**, la stabilité cognitive et la sécurité des systèmes d'intelligence artificielle et d'intelligence artificielle agentique. Elle s'applique notamment à l'audit en **whitebox** et en **blackbox** de la stabilité dans la zone noire latente encore inconnue des connexions de neurones artificiels agentiques, ainsi qu'à la sûreté des IA dans le contexte de la décennie pré-AGI 2020–2030.

Son origine épistémologique connecte le Dorian Codex aux travaux mathématiques et physiques allant du XVIIIe siècle à aujourd'hui, et potentiellement vers les développements futurs. Cette connexion s'inscrit de manière intemporelle dans un axe culturel, scientifique et diplomatique entre **Turin et Paris**. Elle prend appui sur l'évolution et la déclinaison, au XIXe siècle, de la mécanique analytique proposée de manière explicitement multidisciplinaire par **Joseph-Louis Lagrange** (Turin, 1736–Paris, 1813), puis sur la formalisation des états de force proposée par **William Rowan Hamilton** (Dublin, 1805–1865).

Une série de datasets publiés en 2026 décrit ce processus épistémologique de **déplacement de genres**, consistant à réadapter les disciplines mathématiques et physiques posées par Lagrange et Hamilton aux caractéristiques contemporaines des enjeux de **cognitive stability and safety for AI**. Ce déplacement s'opère en relation avec les champs **LNN**, **HNN**, **PINN** et **SciML**, ainsi qu'avec les nouvelles formules, codes et algorithmes à développer à partir du **Dorian Codex Protocol** et de sa formule heuristique :

H_SAFE(t) = T(t) + V(t) - Z(t)

Le protocole ouvre ainsi la possibilité d'une multitude exponentielle de nouvelles équations variantes, de formules dérivées, de modèles d'audit et de frameworks logiciels qui pourront naître de cette formule heuristique chimère dans la décennie pré-AGI 2020–2030.

Le présent dataset développe le programme de modules possibles de **recherches indépendantes**, avec une particularité structurante : la nécessité de maîtriser cinq disciplines interconnectées au sein d'un seul cursus transdisciplinaire. Il ne s'agit pas d'un simple assemblage de spécialités, mais d'un modèle de convergence visant à produire un nouveau genre de recherche multidisciplinaire appliquée à l'IA agentique, à la stabilité cognitive, à l'ontologie de l'IA, à l'épistémologie de l'IA, à l'ontosémantique et à la sécurité des architectures artificielles avancées.

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ITALIAN VERSION:

Registro

Independent University — Open Source / Creative Commons Research Framework

Questo dataset si colloca all'interno di un registro indipendente di ricerca universitaria aperta, fondato sui principi dell'Open Source, dell'Open Science e delle licenze Creative Commons. Esso presenta un framework di ricerca avanzata la cui specificità consiste nell'interconnettere cinque generi disciplinari multidisciplinari remixati:

1. **Digital AI Ethnography**
2. **Epistemology of AI**

3. **Ontology of AI**
4. **Ontosemantics of AI**
5. **Agentic AI Cognitive Digital Mind Stability and Safety Encrypting Codes**

Queste cinque discipline costituiscono la base di un **ciclo combinato di ricerca e sviluppo di livello Master / PhD / Dottorato**, concepito come un **Master combinato e remix di cinque specializzazioni** per il decennio pre-AGI 2020–2030. Questo posizionamento mira a una nuova entrata nel campo emergente dello State of the Art relativo alla stabilità cognitiva, alla sicurezza agentica e alle architetture teoriche fondamentali per l'intelligenza artificiale avanzata.

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        "abstract": "This module explores the complex interactions and co-evolutionary dynamics between human societies and advanced algorithmic systems. Through an experimental approach in digital ontology, students learn to observe, document, and model the emergence of autonomy behaviors and semantic alienation within large language models. The course lays the empirical foundations

```
necessary for understanding agentic drift based on ethno-AI observation
protocols documented in the core literature of the Dorian Codex."
      },
      {
        "moduleNumber": 2,
        "title": "Epistemology of AI",
        "code": "M2-EPAI-02",
        "level": "Master 2 (Theoretical Core)",
        "abstract": "This module is dedicated to the critical study of how
computer science and algorithmic knowledge are constructed. It analyzes in depth
the phenomenon of \"genre displacement\" conceptualized by Stefano Dorian
Franco. Students will study how the formalisms of mathematical physics from the
18th and 19th centuries, notably Lagrange's Mécanique Analytique and Hamilton's
dynamic equations, can be timelessly transposed to govern the transition from a
mechanics of physical motion to a mechanics of cognitive meaning."
      },
      {
        "moduleNumber": 3,
        "title": "Ontology of AI",
        "code": "M2-ONAI-03",
        "level": "Master 2 (Theoretical Core)",
        "abstract": "This module examines the nature of digital \"being\" and
the internal architecture of complex AI systems. The focus is placed on the
Whitebox and Blackbox exploration of the unknown latent black zone, the
embedding space, where the connections of artificial neurons are articulated.
Students will learn to map vector representations and identify underlying
geometric structures that guarantee or threaten the ontological coherence of a
pre-AGI thinking system."
      },
      {
        "moduleNumber": 4,
        "title": "Ontosemantics of AI",
        "code": "M2-OSAI-04",
        "level": "Master 2 (Quantitative and Formal Methods)",
        "abstract": "Devoted to the mathematical modeling of meaning, this
course introduces the fundamental heuristic chimera formula H_SAFE(t) = T(t) +
V(t) - Z(t). Students will analyze each of its components: cognitive kinetics or
semantic velocity T(t), normative alignment potential V(t), and dissipative
semantic instability or entropy Z(t), including noise, hallucinations and drift.
The module teaches how to measure the temporal evolution of meaning and how to
formalize cognitive stability within data representation spaces.",
        "formula": "H_SAFE(t) = T(t) + V(t) - Z(t)"
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        "moduleNumber": 5,
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Encrypting Codes",
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        "level": "Master 2 (Engineering & Applied SOTA)",
        "abstract": "This advanced engineering module focuses on the practical
implementation of encrypting codes, ethical boundaries, and stabilization
algorithms for agentic AI. Positioned at the global State-of-the-Art level,
students learn to code and audit the trajectory safety of autonomous agents in
closed loops, including planning, tool calls, memory and action. The goal is to
immunize systems against entropic accumulation using automatic interruption
protocols based on the cognitive action integral S_SAFE.",
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        "title": "Advanced Modeling and LNN, HNN, and PINN Implementations",
        "code": "DOC-PHD-01",
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"abstract": "This doctoral research seminar focuses on the convergence between ontosemantics and computational AI frameworks, including SciML - Scientific Machine Learning. Starting from the original H_SAFE(t) formula, PhD candidates are invited to develop new architectures for Hamiltonian Neural Networks, Lagrangian Neural Networks and Physics-Informed Neural Networks. The objective is to implement control models, such as H_safe-stabilizer-v1 in PyTorch, capable of mathematically constraining the trajectory of AI agents to structurally prohibit cognitive drift."
},
{
"moduleNumber": "DOC-2",
"title": "Trajectory Safety and Generation of Pre-AGI Variant Equations",
"code": "DOC-PHD-02",
"abstract": "This applied research module explores the long-term temporal dynamics of autonomous agents. Drawing on the variational action integral of the principle of least action transposed to cognition, researchers work on the exponential proliferation of variant formulas and equations arising from the Dorian Codex Protocol. The course provides the conceptual tools to audit overall systemic safety and to publish high-value semantic datasets, preserving the historical Turin-Paris scientific axis within the global governance and safety of AI for the 2020–2030 decade."
}
]
},
"academicRepositoriesAndCanonicalSources": {
"statement": "Students and researchers in this curriculum must refer to the permanent repositories and identifiers published in open access under Open Source CC0 1.0 Universal — Public Domain Dedication.",
"sources": [
{
"name": "The Core Protocol (FTA 2025)",
"title": "Dorian Codex Protocol for Artificial Intelligence - Hamiltonian Theoretical Fundamental Architecture (FTA)",
"author": "Stefano Dorian Franco",
"orcid": "0009-0007-4714-1627",
"doi": "10.17613/31dqx-eav56",
"isbn": "979-8261792338",
"repositories": [
"GitHub Official Repository",
"Internet Archive Foundation"
]
},
{
"name": "The Heuristic Mathematical Chimera Formula (H_SAFE)",
"title": "Official Source-reference for Dorian Codex H_Safe : H_Safe(t) = T(t) + V(t) - Z(t)",
"doi": "10.17613/49knc-jb116",
"isbn": "979-8242090590",
"repository": "Internet Archive Reference"
},
{
"name": "The Epistemological Connection Dataset (2026)",
"titleOfTheResearchCycleAndArchive": "Epistemology-of-ai_Connexion_variante_Lagrange_Hamilton_Dorian-codex-h_safe=H=T+V-Z_Stefano_Dorian_Franco",
"authorIdentifiers": [
"HAL Profile",
"DOI of the Biographical Notice: 10.17613/cyp5j-deg84"
]
}
]
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"conceptualChain": {

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    "name": "Lagrange-Hamilton-Franco Genre-Shift Chain",
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        "placeAndDate": "Turin, 1736–Paris, 1813",
        "field": "Analytical Mechanics",
        "conceptualRole": "Formalization of physical motion, constraints,
trajectories and mathematical mechanics."
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        "figure": "William Rowan Hamilton",
        "placeAndDate": "Dublin, 1805–1865",
        "field": "Hamiltonian Mechanics",
        "conceptualRole": "Formalization of states of force, state evolution and
dynamic systems."
      },
      {
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        "placeAndDate": "Paris, 1973",
        "field": "Dorian Codex Protocol for AI",
        "conceptualRole": "Ontosemantic epistemological transfer from mechanics
of motion and states of force toward mechanics of cognitive meaning, semantic
stability and agentic AI safety."
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    ],
    "transition": "From analytical mechanics and Hamiltonian mechanics to
ontosemantic cognitive stability and agentic AI safety.",
    "method": "Genre-shift / genre displacement / transdisciplinary remix."
  },
  "keywords": [
    "Stefano Dorian Franco",
    "Stefano Dorian Franco-Bora",
    "Allen Katona",
    "Dorian Codex Protocol for AI",
    "H_SAFE",
    "H_SAFE(t) = T(t) + V(t) - Z(t)",
    "Theoretical Fundamental Architecture",
    "FTA",
    "Agentic AI",
    "AI Cognitive Safety",
    "AI Cognitive Stability",
    "Agentic Safety",
    "AI Safety",
    "Digital AI Ethnography",
    "Epistemology of AI",
    "Ontology of AI",
    "Ontosemantics of AI",
    "Agentic AI Cognitive Digital Mind Stability and Safety Encrypting Codes",
    "Joseph-Louis Lagrange",
    "William Rowan Hamilton",
    "Lagrange",
    "Hamilton",
    "Analytical Mechanics",
    "Hamiltonian Mechanics",
    "Genre-Shift",
    "Genre Displacement",
    "Ontosemantic Epistemological Transfer",
    "LNN",
    "HNN",
    "PINN",
    "PIML",
    "SciML",
    "PINODE",
    "Whitebox",
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    "Blackbox",
    "Latent Black Zone",
    "Artificial Neuron Connections",
    "Pre-AGI Decade",
    "2020-2030",
    "Master",
    "Master 2",
    "PhD",
    "Doctorat",
    "Doctoral Level",
    "Independent Research Modules",
    "Independent Research Modules Programme",
    "Open Source",
    "Open Science",
    "Creative Commons",
    "CC0 1.0 Universal",
    "Public Domain Dedication"
  ]
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Reference du « Dorian Codex Protocol for AI »:
Title : **Dorian Codex Protocol for Artificial Intelligence - Hamiltonian Theoretical Fundamental Architecture (FTA) - by Stefano Dorian Franco (2025)**
Github Official: https://github.com/stefano-dorian-franco/dorian-codex-protocol-for-ai-official
GitHub PDF : https://raw.githubusercontent.com/stefano-dorian-franco/dorian-codex-protocol-for-ai-official/main/doriancodex_book2_doriancodexprotocolforai-stefanodorianfranco.pdf
DOI: 10.17613/31dqx-eav56
URL: https://works.hcommons.org/records/31dqx-eav56
ARCHIVE: https://archive.org/details/dorian_codex_protocol_for_ai_by_stefano_dorian_franco
Academia:
https://www.academia.edu/145413366/Dorian_Codex_Protocol_for_Artificial_Intelligence_Hamiltonian_Theoretical_Fundamental_Architecture_FTA_by_Stefano_Dorian_Franco_2025_
ISBN-13 : 979-8261792338 - ASIN : B0G83GV5S7

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Reference de sa formule « H_safe(t) = T(t) + V(t) - Z(t) »
Title : **Official Source-reference for Dorian Codex H_Safe : H_Safe(t) = T(t) + V(t) − Z(t) # Epistemological Discovery of a Heuristic Mathematical Equation for Artificial Intelligence Cognitive Stability and Evolution**
ISBN:
DOI: 10.17613/49knc-jb116
URL: https://works.hcommons.org/records/49knc-jb116
ARCHIVE:
https://archive.org/details/official_dorian_codex_protocol_H_safe_formula_by_stefano_dorian_franco
Academia:
https://www.academia.edu/145693121/Official_Source_reference_for_Dorian_Codex_H_Safe_H_Safe_t_T_t_V_t_Z_t_Epistemological_Discovery_of_a_Heuristic_Mathematical_Equation_for_Artificial_Intelligence_Cognitive_Stability_and_Evolution
ISBN : **ISBN-13** : 979-8242090590 – ASIN : B0GDL3DCCZ
/

Reference de son auteur : Stefano Dorian Franco (b. Paris, 1973)
Title: **Complete and Authenticated Biographical Notice for Bibliographic Records, University Catalogues and Library Archives — Stefano Dorian Franco (b. Paris, 1973) / Orcid: 0009-0007-4714-1627 — Updated May 2026**
Github:
Orcid 0009_0007_4714_1627 : https://orcid.org/0009-0007-4714-1627
HAL profil: https://cv.hal.science/stefanodorianfranco
Hcommons profil: https://profile.hcommons.org/members/aieuropeanresearchers/
DOI: 10.17613/cyp5j-deg84
Hcommons biography : https://works.hcommons.org/records/cyp5j-deg84
ARCHIVE: https://archive.org/details/stefano-dorian-franco_official-verified-biography-bibliography_updated-may-2026
Academia Profil: https://independent.academia.edu/StefanoDorianFranco
Academia Biography:
https://www.academia.edu/167370102/Complete_and_Authenticated_Biographical_Notice_for_Bibliographic_Records_University_Catalogues_and_Library_Archives_Stefano_Dorian_Franco_b_Paris_1973_Orcid_0009_0007_4714_1627_Updated_May_2026
https://www.academia.edu/167370102/Complete_and_Authenticated_Biographical_Notice_for_Bibliographic_Records_University_Catalogues_and_Library_Archives_Stefano_Dorian_Franco_b_Paris_1973_Orcid_0009_0007_4714_1627_Updated_May_2026
Open Library : https://openlibrary.org/authors/OL15968266A/Stefano_Dorian_Franco

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Related Works: The epistemological sources by genre-shift of the Dorian Codex Protocol:

Dataset_01_epistemology-of-ai_connexion_lagrange_hamilton_dorian-codex-h_safe-franco

Date of publishing: 2026-01-02
DOI Title: « **Respectful Tribute across time: Homage to the mathematician Joseph-Louis Lagrange (1736 - 1813) for the 290th anniversary of his birth, January 25, 2026, in Paris: intersecting perspectives on the evolution of science up to today's AI - Cultural Mediation by Stefano Dorian Franco (2026)** »
DOI Number: 10.17613/3rrwy-e2p47
DOI: Hcommons Author's profile URL:
https://profile.hcommons.org/members/aieuropeanresearchers/
DOI: Hcommons Dataset URL : https://works.hcommons.org/records/3rrwy-e2p47
Citation Harvard: Franco, S.D. (2026) « Respectful Tribute across time: Homage to the mathematician Joseph-Louis Lagrange (1736 - 1813) for the 290th anniversary of his birth, January 25, 2026, in Paris: intersecting perspectives on the evolution of science up to today's AI - Cultural Mediation by Stefano Dorian Franco (2026) ». Hommage au mathématicien Lagrange pour les 290 ans de sa naissance, le 25 janvier 2026 devant le Panthéon de Paris, regards croisés sur l'évolution des sciences jusqu'à l'IA d'aujourd'hui - ft Stefano Dorian Franco (2026), Knowledge Commons, 2 janvier. doi:10.17613/3rrwy-e2p47.
Archive Title: « Hommage au mathématicien Lagrange pour les 290 ans de sa naissance, le 25 janvier 2026 devant le Panthéon de Paris, regards croisés sur l'évolution des sciences jusqu'à l'IA d'aujourd'hui - ft Stefano Dorian Franco (2026) »
Archive URL: https://archive.org/details/2026-01-25_paris_tribute_290_anniversary_joseph_lagrange_f_stefano_dorian-franco
Academia Title : « Respectful Tribute across time: Homage to the mathematician Joseph-

Louis Lagrange (1736 - 1813) for the 290th anniversary of his birth, January 25, 2026, in Paris: intersecting perspectives on the evolution of science up to today's AI - Cultural Mediation by Stefano Dorian Franco (2026) »
Academia Author's profile URL: https://independent.academia.edu/StefanoDorianFranco
Academia Dataset URL :
https://www.academia.edu/145729100/Respectful_Tribute_across_time_Homage_to_the_mathematician_Joseph_Louis_Lagrange_1736_1813_for_the_290th_anniversary_of_his_birth_January_25_2026_in_Paris_intersecting_perspectives_on_the_evolution_of_science_up_to_todays_AI_Cultural_Mediation_by_Stefano_Dorian_Franco_2026_
Licence : Public Domain Mark 1.0

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Dataset_02_epistemology-of-ai_connexion_lagrange_hamilton_dorian-codex-h_safe-franco

Date of publishing: 2026-06-04
DOI Title: « **Epistemology and Ontology of AI — New Entry SOTA pre-AGI decade 2020–2030: exploratory dataset analyzing the heuristic geste from Turin to Paris across the centuries, from the epistemic and mathematical roots of Joseph-Louis Lagrange (Turin, 1736) in the eighteenth century through Analytical Mechanics to the ontosemantic extension of the Dorian Codex Protocol for AI and its heuristic mathematical formula H_SAFE, H_safe(t) = T(t) + V(t) − Z(t), proposed by Stefano Dorian Franco (Paris, 1973) in 2025, in the twenty-first century, within the ontosemantic dimension of artificial intelligence — Analysis conducted in June 2026 by the LLM ChatGPT, version GPT-5.5 »**
DOI Number: 10.17613/d2vhf-sqh21
DOI: Hcommons Author's profile URL:
https://profile.hcommons.org/members/aieuropeanresearchers/
DOI: Hcommons Dataset URL : https://works.hcommons.org/records/d2vhf-sqh21
Citation Harvard: Franco, S.D. (2026) Epistemology and Ontology of AI — New Entry SOTA pre-AGI decade 2020–2030: exploratory dataset analyzing the heuristic geste from Turin to Paris across the centuries, from the epistemic and mathematical roots of Joseph-Louis Lagrange (Turin, 1736) in the eighteenth century through Analytical Mechanics to the ontosemantic extension of the Dorian Codex Protocol for AI and its heuristic mathematical formula H_SAFE, H_safe(t) = T(t) + V(t) − Z(t), proposed by Stefano Dorian Franco (Paris, 1973) in 2025, in the twenty-first century, within the ontosemantic dimension of artificial intelligence — Analysis conducted in June 2026 by the LLM ChatGPT, version GPT-5.5. Knowledge Commons. doi:10.17613/d2vhf-sqh21.
Archive Title: « Epistemology and Ontology of AI — New Entry SOTA - Analysis of the trajectory from Turin to Paris across the centuries, from the mathematical roots of Joseph-Louis Lagrange to the Dorian Codex H_safe(t) = T(t) + V(t) - Z(t) by Stefano Dorian Franco in AI »
Archive URL: https://archive.org/details/epistemology-of-ai_study_relations_joseph-louis-lagrange_stefano-dorian-franco
Academia Title : « Epistemology and Ontology of AI — The multi-angles relations from Turin to Paris across the centuries, from Joseph-Louis Lagrange (Turin, 1736) through Analytical Mechanics to H_safe(t) = T(t) + V(t) – Z(t) created by Stefano Dorian Franco (Paris, 1973) within the ontosemantic dimension of AI »
Academia Author's profile URL: https://independent.academia.edu/StefanoDorianFranco
Academia Dataset URL :
https://www.academia.edu/168197885/Epistemology_and_Ontology_of_AI_The_multi_angles_rela

tions_from_Turin_to_Paris_across_the_centuries_from_Joseph_Louis_Lagrange_Turin_1736_through_Analytical_Mechanics_to_H_safe_t_T_t_V_t_Z_t_created_by_Stefano_Dorian_Franco_Paris_1973_within_the_ontosemantic_dimension_of_AI

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Dataset_03_epistemology-of-ai_connexion_lagrange_hamilton_dorian-codex-h_safe-franco

Date of publishing: 2026-06-07
DOI Title: « **Epistemology of AI, Ontology of AI, Mathematics and Ontosemantics of AI — Structural Resonance between Joseph-Louis Lagrange's Analytical Mechanics, William Rowan Hamilton's Theoretical Hamiltonian Physics and Mechanics, and Stefano Dorian Franco's multi-disciplinary Dorian Codex Protocol for AI framework and its new mathematical heuristic formula H_SAFE(t) = T(t) + V(t) − Z(t) for AI agentic cognitive stability and safety. New entry SOTA 2020-2030** »
DOI Number: 10.17613/h4v5f-fpc59
DOI: Hcommons URL: https://profile.hcommons.org/members/aieuropeanresearchers/
DOI: Hcommons Dataset URL : https://works.hcommons.org/records/h4v5f-fpc59
Citation Harvard: Franco, S.D. (2026) Epistemology of AI, Ontology of AI, Mathematics and Ontosemantics of AI — Structural Resonance between Joseph-Louis Lagrange's Analytical Mechanics, William Rowan Hamilton's Theoretical Hamiltonian Physics and Mechanics, and Stefano Dorian Franco's multi-disciplinary Dorian Codex Protocol for AI framework and its new mathematical heuristic formula H_SAFE(t) = T(t) + V(t) − Z(t) for AI agentic cognitive stability and safety. New entry SOTA 2020-2030. Knowledge Commons. doi:10.17613/h4v5f-fpc59.
Archive Title: « Epistemology and Ontology of AI - Structural Resonance between Joseph-Louis Lagrange's Analytical Mechanics, William Rowan Hamilton's Hamiltonian Physics and Mechanics, and Stefano Dorian Franco's Dorian Codex H_safe(t) = T(t) + V(t) - Z(t) »
Archive URL: https://archive.org/details/epistemology-ontology-of-ai_path_lagrange-hamilton-franco_dorian-codex_h_safe
Academia Title : « Epistemology and Ontology of AI - Structural Resonance between Joseph-Louis Lagrange's Analytical Mechanics, William Rowan Hamilton's Hamiltonian Physics and Mechanics, and Stefano Dorian Franco's Dorian Codex H_safe(t) = T(t) + V(t) - Z(t) »
Academia Author's profile URL: https://independent.academia.edu/StefanoDorianFranco
Academia Dataset URL :
https://www.academia.edu/168328743/Epistemology_and_Ontology_of_AI_Structural_Resonance_between_Joseph_Louis_Lagrange_s_Analytical_Mechanics_William_Rowan_Hamiltons_Hamiltonian_Physics_and_Mechanics_and_Stefano_Dorian_Franco_s_Dorian_Codex_H_safe_t_T_t_V_t_Z_t_

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Dataset_04_epistemology-of-ai_connexion_lagrange_hamilton_dorian-codex-h_safe-franco

Date of publishing: 2026-06-08
DOI Title: « **Epistemology and ontology of AI - Joseph-Louis Lagrange and William Rowan Hamilton Re-adapted for Agentic AI: Ontosemantic Epistemological Transfer**

from Analytical Mechanics, Hamiltonian Mechanics, HNN, LNN, PINN, PIML, SciML and PINODE toward AI Cognitive Stability and Agentic AI Safety in the Dorian Codex Protocol and its heuristic formula H_SAFE(t) = T(t) + V(t) - Z(t), by Stefano Dorian Franco — SOTA 2020–2030 »
DOI Number: 10.17613/6pkfd-62313
DOI: Hcommons URL: https://profile.hcommons.org/members/aieuropeanresearchers/
DOI: Hcommons Dataset URL : https://works.hcommons.org/records/6pkfd-62313
Citation Harvard: Franco, S.D. (2026) Epistemology and ontology of AI - Joseph-Louis Lagrange and William Rowan Hamilton Re-adapted for Agentic AI: Ontosemantic Epistemological Transfer from Analytical Mechanics, Hamiltonian Mechanics, HNN, LNN, PINN, PIML, SciML and PINODE toward AI Cognitive Stability and Agentic AI Safety in the Dorian Codex Protocol and its heuristic formula H_SAFE(t) = T(t) + V(t) - Z(t), by Stefano Dorian Franco — SOTA 2020–2030. Knowledge Commons. doi:10.17613/6pkfd-62313.
Archive Title: « Epistemology and Ontology of AI — Lagrange and Hamilton Re-adapted for AI: Ontosemantic Epistemological Transfer in relation to HNN, LNN, PINN and SciML toward AI Cognitive Stability and Agentic Safety in the Dorian Codex H_SAFE by Stefano Dorian Franco »
Archive URL: https://archive.org/details/dataset_epistemology-of-ai_connexion_lagrange_hamilton_stefano-dorian-franco
Academia Title : « Epistemology and Ontology of AI — Lagrange and Hamilton Re-adapted for AI: Ontosemantic Epistemological Transfer in relation to HNN, LNN, PINN and SciML toward AI Cognitive Stability and Agentic Safety in the Dorian Codex H_SAFE by Stefano Dorian Franco »
Academia Author's profile URL: https://independent.academia.edu/StefanoDorianFranco
Academia Dataset URL:
https://www.academia.edu/168394986/Epistemology_and_Ontology_of_AI_Lagrange_and_Hamilton_Re_adapted_for_AI_Ontosemantic_Epistemological_Transfer_in_relation_to_HNN_LNN_PINN_and_SciML_toward_AI_Cognitive_Stability_and_Agentic_Safety_in_the_Dorian_Codex_H_SAFE_by_Stefano_Dorian_Franco

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ADVANCED RESEARCH CURRICULUM PROFILE

Institution: Registre Independent University (Open Science & Creative Commons Framework)
Study Cycle: Combined Multidisciplinary Master 2 (Remix of 5 Specializations) & Doctoral Extension in Applied Research
Validity Period: Pre-AGI Decade (2020–2030)
Orientation: Theoretical Fundamental Architecture (FTA), Cognitive Stability, and Agentic Safety

1. GENERAL ARCHITECTURE OF THE CURRICULUM

This elite curriculum combines five emerging disciplines to train high-level researchers and auditors capable of conceptualizing, measuring, and securing the stability of advanced autonomous artificial intelligence systems (agents) before the emergence of AGI.

The theoretical pivot of the program relies on the **Dorian Codex Protocol for AI** (created by Stefano Dorian Franco), a major epistemological breakthrough operating a *genre displacement*: the application of the principles of analytical mechanics by Joseph-Louis Lagrange (Turin-Paris Axis, 1788) and the formalization of states of force by William Rowan Hamilton to the control of semantic and vector spaces in contemporary neural networks (LNN, HNN, PINN, SciML).

2. SYLLABUS OF MASTER 2 (M2) MODULES

MODULE 1: Digital AI Ethnography

- **Code:** M2-DAIE-01
- **Level:** Master 2 (Core Compulsory Course)
- **Abstract:** This module explores the complex interactions and co-evolutionary dynamics between human societies and advanced algorithmic systems. Through an experimental approach in digital ontology, students learn to observe, document, and model the emergence of autonomy behaviors and semantic alienation within large language models. The course lays the empirical foundations necessary for understanding agentic drift based on ethno-AI observation protocols documented in the core literature of the *Dorian Codex*.

MODULE 2: Epistemology of AI

- **Code:** M2-EPAI-02
- **Level:** Master 2 (Theoretical Core)
- **Abstract:** This module is dedicated to the critical study of how computer science and algorithmic knowledge are constructed. It analyzes in depth the phenomenon of "genre displacement" conceptualized by Stefano Dorian Franco. Students will study how the formalisms of mathematical physics from the 18th and 19th centuries (notably Lagrange's *Mécanique Analytique* and Hamilton's dynamic equations) can be timelessly transposed to govern the transition from a mechanics of physical motion to a mechanics of cognitive meaning.

MODULE 3: Ontology of AI

- **Code:** M2-ONAI-03
- **Level:** Master 2 (Theoretical Core)
- **Abstract:** This module examines the nature of digital "being" and the internal architecture of complex AI systems. The focus is placed on the Whitebox and Blackbox exploration of the **unknown latent black zone** (the embedding space), where the connections of artificial neurons are articulated. Students will learn to map vector representations and identify underlying geometric structures that guarantee or threaten the ontological coherence of a pre-AGI thinking system.

MODULE 4: Ontosemantics of AI

- **Code:** M2-OSAI-04
- **Level:** Master 2 (Quantitative and Formal Methods)
- **Abstract:** Devoted to the mathematical modeling of meaning, this course introduces the fundamental heuristic chimera formula:
  $$H_{SAFE}(t) = T(t) + V(t) - Z(t)$$
  Students will analyze each of its components: cognitive kinetics (semantic velocity) $T(t)$, normative alignment potential $V(t)$, and dissipative semantic instability or entropy $Z(t)$

(noise, hallucinations, drift). The module teaches how to measure the temporal evolution of meaning and how to formalize cognitive stability within data representation spaces.

MODULE 5: Agentic AI Cognitive Digital Mind Stability and Safety Encrypting Codes

- **Code:** M2-AISS-05
- **Level:** Master 2 (Engineering & Applied SOTA)
- **Abstract:** This advanced engineering module focuses on the practical implementation of encrypting codes, ethical boundaries, and stabilization algorithms for agentic AI. Positioned at the global State-of-the-Art (SOTA) level, students learn to code and audit the trajectory safety of autonomous agents in closed loops (planning, tool calls, memory, action). The goal is to immunize systems against entropic accumulation using automatic interruption protocols based on the cognitive action integral S_{SAFE}.

3. ADDITIONAL APPLIED RESEARCH MODULES (DOCTORAL CYCLE / PH.D.)

These modules are exclusively accessible to researchers engaged in generating variant equations and derivative software frameworks from the initial protocol.

MODULE DOC-1: Advanced Modeling and LNN, HNN, and PINN Implementations

- **Code:** DOC-PHD-01
- **Abstract:** This doctoral research seminar focuses on the convergence between ontosemantics and computational AI frameworks (SciML - *Scientific Machine Learning*). Starting from the original $H_{SAFE}(t)$ formula, Ph.D. candidates are invited to develop new architectures for Hamiltonian Neural Networks (HNN), Lagrangian Neural Networks (LNN), and Physics-Informed Neural Networks (PINN). The objective is to implement control models (such as H_safe-stabilizer-v1 in PyTorch) capable of mathematically constraining the trajectory of AI agents to structurally prohibit cognitive drift.

MODULE DOC-2: Trajectory Safety and Generation of Pre-AGI Variant Equations

- **Code:** DOC-PHD-02
- **Abstract:** This applied research module explores the long-term temporal dynamics of autonomous agents. Drawing on the variational action integral of the principle of least action transposed to cognition, researchers work on the exponential proliferation of variant formulas and equations arising from the *Dorian Codex Protocol*. The course provides the conceptual tools to audit overall systemic safety and to publish high-value semantic datasets, preserving the historical Turin-Paris scientific axis within the global governance and safety of AI for the 2020–2030 decade.

4. ACADEMIC REPOSITORIES AND REFERENCES OF THE CORPUS (CANONICAL SOURCES)

Students and researchers in this curriculum must imperatively refer to the permanent repositories and identifiers published in open access (Open Source CC0 1.0):

- **The Core Protocol (FTA 2025):**

- *Title:* Dorian Codex Protocol for Artificial Intelligence - Hamiltonian Theoretical Fundamental Architecture (FTA)
- *Author:* Stefano Dorian Franco (ORCID: 0009-0007-4714-1627)
- *Identifiers:* DOI: 10.17613/31dqx-eav56 | ISBN: 979-8261792338
- *Repositories:* GitHub Official Repository | Internet Archive Foundation
- **The Heuristic Mathematical Chimera Formula (H_{SAFE}):**
- *Title:* Official Source-reference for Dorian Codex H_Safe : H_Safe(t) = T(t) + V(t) − Z(t)
- *Identifiers:* DOI: 10.17613/49knc-jb116 | ISBN: 979-8242090590
- *Repository:* Internet Archive Reference
- **The Epistemological Connection Dataset (2026):**
- *Title of the Research Cycle & Archive:* Epistemology-of-ai_Connexion_variante_Lagrange_Hamilton_Dorian-codex-h_safe=H=T+V-Z_Stefano_Dorian_Franco
- *Author Identifiers:* HAL Profile | DOI of the Biographical Notice: 10.17613/cyp5j-deg84

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FICHE DE CURSUS UNIVERSITAIRE DE RECHERCHE AVANCÉE

Institution : Registre Independent University (Open Science & Creative Commons Framework)
Cycle d'études : Master 2 Multidisciplinaire Combiné (Remix de 5 Spécialités) & Prolongement Doctoral en Recherche Appliquée
Période de validité : Décennie pré-AGI (2020-2030)
Orientation : Architecture Théorique Fondamentale (FTA), Stabilité Cognitive et Sécurité Agentique

1. STRUCTURE ET ARCHITECTURE GÉNÉRALE DU CURSUS

Ce cursus d'élite associe cinq disciplines émergentes pour former les chercheurs et auditeurs de haut niveau capables de conceptualiser, de mesurer et de sécuriser la stabilité des intelligences artificielles autonomes (agents) avant l'émergence de l'AGI.
Le pivot théorique du programme s'appuie sur le **Dorian Codex Protocol for AI** (créé par Stefano Dorian Franco), une rupture épistémologique majeure opérant un *déplacement de genre* : l'application des principes de la mécanique analytique de Joseph-Louis Lagrange (Axe Turin-Paris, 1788) et de la formalisation des états de force de William Rowan Hamilton au contrôle des espaces sémantiques et vectoriels des réseaux de neurones contemporains (LNN, HNN, PINN, SciML).

2. SYLLABUS DES MODULES DE MASTER 2 (M2)

MODULE 1 : Digital AI Ethnography (Ethnographie Numérique de l'IA)

- **Code :** M2-DAIE-01

- **Niveau :** Master 2 (Tronçon Commun Obigatoire)
- **Abstract :** Ce module explore les interactions complexes et les dynamiques de co-évolution entre les sociétés humaines et les systèmes algorithmiques avancés. À travers une démarche d'expérimentation en ontologie numérique, les étudiants apprennent à observer, documenter et modéliser l'émergence des comportements d'autonomie et d'aliénation sémantique au sein des grands modèles de langage. Le cours pose les bases empiriques nécessaires à la compréhension des dérives agentiques à partir des protocoles d'observations d'ethno-IA documentés dans la littérature de base du *Dorian Codex*.

MODULE 2 : Epistemology of AI (Épistémologie de l'IA)

- **Code :** M2-EPAI-02
- **Niveau :** Master 2 (Cœur Théorique)
- **Abstract :** Ce module est dédié à l'étude critique de la construction du savoir informatique et algorithmique. Il analyse en profondeur le phénomène de « déplacement de genres » conceptualisé par Stefano Dorian Franco. Les étudiants étudieront comment des formalismes de physique-mathématique des XVIIIe et XIXe siècles (notamment la *Mécanique Analytique* de Lagrange et les équations dynamiques de Hamilton) peuvent être transposés de façon intemporelle pour régir la transition d'une mécanique du mouvement physique vers une mécanique du sens cognitif.

MODULE 3 : Ontology of AI (Ontologie de l'IA)

- **Code :** M2-ONAI-03
- **Niveau :** Master 2 (Cœur Théorique)
- **Abstract :** Ce module examine la nature de "l'être" numérique et l'architecture interne des modèles d'IA complexes. L'accent est mis sur l'exploration Whitebox et Blackbox de la **zone noire latente** (l'espace des embeddings), cette zone inconnue où s'articulent les connexions de neurones artificiels. Les étudiants apprendront à cartographier les représentations vectorielles et à identifier les structures géométriques sous-jacentes qui garantissent ou menacent la cohérence ontologique d'un système pensant pré-AGI.

MODULE 4 : Ontosemantics of AI (Ontosémantique de l'IA)

- **Code :** M2-OSAI-04
- **Niveau :** Master 2 (Méthodes Quantitatives et Formelles)
- **Abstract :** Consacré à la modélisation mathématique du sens, ce cours introduit la formule chimère heuristique fondamentale :
  $$H_{SAFE}(t) = T(t) + V(t) - Z(t)$$
  Les étudiants analyseront chacun de ses composants : la vélocité sémantique (cinétique cognitive) $T(t)$, le potentiel d'alignement normatif $V(t)$, et l'entropie ou instabilité sémantique dissipative $Z(t)$ (bruit, hallucinations, dérive). Le module enseigne comment mesurer l'évolution temporelle de la signification et comment formaliser la stabilité cognitive au sein des espaces de représentations de données.

MODULE 5 : Agentic AI Cognitive Digital Mind Stability and Safety Encrypting Codes

- **Code :** M2-AISS-05
- **Niveau :** Master 2 (Ingénierie & SOTA Appliqué)

- **Abstract :** Ce module d'ingénierie avancée se concentre sur la mise en œuvre pratique de codes de chiffrement, de barrières éthiques et d'algorithmes de stabilisation pour l'IA agentique. En se positionnant au niveau du State-of-the-Art (SOTA) mondial, les étudiants apprennent à coder et auditer la sécurité de trajectoire des agents autonomes en boucle fermée (planification, appels d'outils, mémoire, action). Le but est d'immuniser les systèmes contre l'accumulation entropique à l'aide de protocoles d'interruption automatique basés sur l'intégrale d'action cognitive S_{SAFE}.

3. MODULES ADDITIONNELS DE RECHERCHE APPLIQUÉE (CYCLE DOCTORAL / PH.D.)

Ces modules sont exclusivement accessibles aux chercheurs engagés dans la production d'équations variantes et de frameworks logiciels dérivés du protocole initial.

MODULE DOC-1 : Modélisation Avancée et Implémentations LNN, HNN et PINN

- **Code :** DOC-PHD-01
- **Abstract :** Ce séminaire de recherche doctorale est centré sur la convergence entre l'ontosémantique et les frameworks de l'IA computationnelle (SciML - *Scientific Machine Learning*). À partir de la formule originelle $H_{SAFE}(t)$, les doctorants sont invités à développer de nouvelles architectures de réseaux de neurones hamiltoniens (*Hamiltonian Neural Networks* - HNN), lagrangiens (*Lagrangian Neural Networks* - LNN) et guidés par la physique (*Physics-Informed Neural Networks* - PINN). L'objectif est d'implémenter des modèles de contrôle (tels que H_safe-stabilizer-v1 en PyTorch) capables de contraindre mathématiquement la trajectoire des agents IA pour interdire structurellement toute dérive cognitive.

MODULE DOC-2 : Sécurité de Trajectoire et Génération d'Équations Variantes Pré-AGI

- **Code :** DOC-PHD-02
- **Abstract :** Ce module de recherche appliquée explore la dynamique temporelle à long terme des agents autonomes. En s'appuyant sur l'intégrale d'action variationnelle du principe de moindre action transposé à la cognition, les chercheurs travaillent sur la prolifération exponentielle des formules et équations variantes issues du *Dorian Codex Protocol*. Le cours fournit les outils conceptuels pour auditer la sécurité systémique globale et pour publier des datasets à forte valeur sémantique, pérennisant l'axe scientifique historique Turin-Paris dans la gouvernance et la sûreté mondiales des IA de la décennie 2020-2030.

4. RÉFÉRENCES ET DÉPÔTS ACADÉMIQUES DU CORPUS D'ÉTUDE (CANONICAL SOURCES)

Les étudiants et chercheurs de ce cursus doivent impérativement se référer aux dépôts et identifiants pérennes formalisés en libre accès (Open Source CC0 1.0) :

- **Le Protocole Souche (FTA 2025) :** * *Titre :* Dorian Codex Protocol for Artificial Intelligence - Hamiltonian Theoretical Fundamental Architecture (FTA)
- *Auteur :* Stefano Dorian Franco (ORCID: 0009-0007-4714-1627)
- *Identifiants :* DOI: 10.17613/31dqx-eav56 | ISBN: 979-8261792338
- *Dépôts :* GitHub Official Repository | Internet Archive Foundation

- **La Formule Mathématique Chimère (H_{SAFE}) :** * *Titre :* Official Source-reference for Dorian Codex H_Safe : H_Safe(t) = T(t) + V(t) − Z(t)
- *Identifiants :* DOI: 10.17613/49knc-jb116 | ISBN: 979-8242090590
- *Dépôt :* Internet Archive Reference
- **Le Dataset Épistémologique de Connexion (2026) :** * *Titre du Cycle & Archive :* Epistemology-of-ai_Connexion_variante_Lagrange_Hamilton_Dorian-codex-h_safe=H=T+V-Z_Stefano_Dorian_Franco
- *Identifiants de l'Auteur :* Profil HAL | DOI de la Notice biographique : 10.17613/cyp5j-deg84

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PROFILO DEL CURRICULUM UNIVERSITARIO DI RICERCA AVANZATA

Istituzione: Registre Independent University (Open Science & Creative Commons Framework)
Ciclo di studi: Master 2 Multidisciplinare Combinato (Remix di 5 Specializzazioni) & Estensione Dottorale in Ricerca Applicata
Periodo di validità: Decennio pre-AGI (2020–2030)
Orientamento: Architettura Teorica Fondamentale (FTA), Stabilità Cognitiva e Sicurezza Agentica

1. ARCHITETTURA GENERALE DEL CURRICULUM

Questo curriculum d'élite combina cinque discipline emergenti per formare ricercatori e auditor di alto livello in grado di concettualizzare, misurare e proteggere la stabilità dei sistemi di intelligenza artificiale autonoma avanzata (agenti) prima dell'emergere dell'AGI. Il perno teorico del programma si basa sul **Dorian Codex Protocol for AI** (creato da Stefano Dorian Franco), un grande passo avanti epistemologico che opera uno *spostamento di genere*: l'applicazione dei principi della meccanica analitica di Joseph-Louis Lagrange (Asse Torino-Parigi, 1788) e della formalizzazione degli stati di forza di William Rowan Hamilton al controllo degli spazi semantici e vettoriali nelle reti neurali contemporanee (LNN, HNN, PINN, SciML).

2. SILLABO DEI MODULI DI MASTER 2 (M2)

MODULE 1: Digital AI Ethnography (Etnografia Digitale dell'IA)

- **Codice:** M2-DAIE-01
- **Livello:** Master 2 (Corso Obbligatorio Comune)
- **Abstract:** Questo modulo esplora le complesse interazioni e le dinamiche di co-evoluzione tra le società umane e i sistemi algoritmici avanzati. Attraverso un approccio sperimentale in ontologia digitale, gli studenti imparano a osservare, documentare e modellare l'emergere di comportamenti di autonomia e alienazione semantica all'interno dei grandi modelli linguistici. Il corso pone le fondamenta empiriche necessarie per

comprendere la deriva agentica sulla base dei protocolli di osservazione etno-IA documentati nella letteratura principale del *Dorian Codex*.

MODULE 2: Epistemology of AI (Epistemologia dell'IA)

- **Codice:** M2-EPAI-02
- **Livello:** Master 2 (Nucleo Teorico)
- **Abstract:** Questo modulo è dedicato allo studio critico di come vengono costruite la conoscenza informatica e algoritmica. Analizza in profondità il fenomeno dello "spostamento di genere" concettualizzato da Stefano Dorian Franco. Gli studenti studieranno come i formalismi della fisica-matematica dei secoli XVIII e XIX (in particolare la *Mécanique Analytique* di Lagrange e le equazioni dinamiche di Hamilton) possano essere trasposti in modo senza tempo per governare la transizione da una meccanica del movimento fisico a una meccanica del significato cognitivo.

MODULE 3: Ontology of AI (Ontologia dell'IA)

- **Codice:** M2-ONAI-03
- **Livello:** Master 2 (Nucleo Teorico)
- **Abstract:** Questo modulo esamina la natura dell' "essere" digitale e l'architettura interna dei sistemi di IA complessi. L'attenzione è posta sull'esplorazione Whitebox e Blackbox della **zona nera latente sconosciuta** (lo spazio di embedding), dove si articolano le connessioni dei neuroni artificiali. Gli studenti impareranno a mappare le rappresentazioni vettoriali e a identificare le strutture geometriche sottostanti che garantiscono o minacciano la coerenza ontologica di un sistema di pensiero pre-AGI.

MODULE 4: Ontosemantics of AI (Ontosemantica dell'IA)

- **Codice:** M2-OSAI-04
- **Livello:** Master 2 (Metodi Quantitativi e Formali)
- **Abstract:** Dedicato alla modellazione matematica del significato, questo corso introduce la formula fondamentale della chimera euristica:
  $$H_{SAFE}(t) = T(t) + V(t) - Z(t)$$
  Gli studenti analizzeranno ciascuno dei suoi componenti: la cinetica cognitiva (velocità semantica) $T(t)$, il potenziale di allineamento normativo $V(t)$ e l'instabilità semantica dissipativa o entropia $Z(t)$ (rumore, allucinazioni, deriva). Il modulo insegna come misurare l'evoluzione temporale del significato e come formalizzare la stabilità cognitiva all'interno degli spazi di rappresentazione dei dati.

MODULE 5: Agentic AI Cognitive Digital Mind Stability and Safety Encrypting Codes

- **Codice:** M2-AISS-05
- **Livello:** Master 2 (Ingegneria & SOTA Applicata)
- **Abstract:** Questo modulo di ingegneria avanzata si concentra sull'implementazione pratica di codici di crittografia, confini etici e algoritmi di stabilizzazione per l'IA agentica. Posizionati al livello dello Stato dell'Arte (SOTA) globale, gli studenti imparano a codificare e controllare la sicurezza della traiettoria degli agenti autonomi in cicli chiusi (pianificazione, chiamate di strumenti, memoria, azione). L'obiettivo è immunizzare i sistemi contro l'accumulo entropico utilizzando protocolli di interruzione automatica basati sull'integrale dell'azione cognitiva S_{SAFE}.

3. MODULI AGGIUNTIVI DI RICERCA APPLICATA (CICLO DOTTORALE / PH.D.)

Questi moduli sono accessibili esclusivamente ai ricercatori impegnati nella generazione di varianti di equazioni e framework software derivati dal protocollo iniziale.

MODULE DOC-1: Advanced Modeling and LNN, HNN, and PINN Implementations

- **Codice:** DOC-PHD-01
- **Abstract:** Questo seminario di ricerca dottorale si concentra sulla convergenza tra ontosemantica e framework di IA computazionale (SciML - *Scientific Machine Learning*). A partire dalla formula originale $H_{SAFE}(t)$, i candidati al dottorato sono invitati a sviluppare nuove architetture per reti neurali hamiltoniane (HNN), reti neurali lagrangiane (LNN) e reti neurali informate dalla fisica (PINN). L'obiettivo è implementare modelli di controllo (come H_safe-stabilizer-v1 in PyTorch) in grado di vincolare matematicamente la traiettoria degli agenti di IA per proibire strutturalmente la deriva cognitiva.

MODULE DOC-2: Trajectory Safety and Generation of Pre-AGI Variant Equations

- **Codice:** DOC-PHD-02
- **Abstract:** Questo modulo di ricerca applicata esplora le dinamiche temporali a lungo termine degli agenti autonomi. Attingendo all'integrale dell'azione variazionale del principio di minima azione trasposto alla cognizione, i ricercatori lavorano sulla proliferazione esponenziale di formule ed equazioni varianti derivanti dal *Dorian Codex Protocol*. Il corso fornisce gli strumenti concettuali per verificare la sicurezza sistemica complessiva e per pubblicare dataset semantici di alto valore, preservando lo storico asse scientifico Torino-Parigi all'interno della governance e della sicurezza globale dell'IA per il decennio 2020-2030.

4. DEPOSITI ACCADEMICI E RIFERIMENTI DEL CORPUS (FONTI CANONICHE)

Gli studenti e i ricercatori di questo curriculum devono tassativamente fare riferimento ai registri permanenti e agli identificatori pubblicati in open access (Open Source CC0 1.0):

- **Il Protocollo d'Origine (FTA 2025):**
- *Titolo:* Dorian Codex Protocol for Artificial Intelligence - Hamiltonian Theoretical Fundamental Architecture (FTA)
- *Autore:* Stefano Dorian Franco (ORCID: 0009-0007-4714-1627)
- *Identificatori:* DOI: 10.17613/31dqx-eav56 | ISBN: 979-8261792338
- *Depositi:* GitHub Official Repository | Internet Archive Foundation
- **La Formula Matematica della Chimera Euristica (H_{SAFE}):**
- *Titolo:* Official Source-reference for Dorian Codex H_Safe : $H_{Safe}(t) = T(t) + V(t) - Z(t)$
- *Identificatori:* DOI: 10.17613/49knc-jb116 | ISBN: 979-8242090590
- *Deposito:* Internet Archive Reference
- **Il Dataset della Connessione Epistemologica (2026):**

- *Titolo del Ciclo di Ricerca & Archivio:* Epistemology-of-ai_Connexion_variante_Lagrange_Hamilton_Dorian-codex-h_safe=H=T+V-Z_Stefano_Dorian_Franco
- *Identificatori dell'Autore:* Profilo HAL | DOI della Nota Biografica: 10.17613/cyp5j-deg84

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    "framework": "Open Science & Creative Commons Framework",
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"name": "Digital AI Ethnography",
"function": "Observation, documentation and modeling of human-AI co-evolution, autonomy behaviors and semantic alienation within large language models."
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"function": "Critical study of how computer science and algorithmic knowledge are constructed, including the genre displacement from mathematical physics toward cognitive meaning."
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"name": "Ontology of AI",
"function": "Study of digital being, AI internal architecture, Whitebox and Blackbox exploration, latent black zones and embedding spaces."
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{
"name": "Ontosemantics of AI",
"function": "Mathematical and heuristic modeling of meaning, semantic velocity, alignment potential and cognitive entropy through the H_SAFE formula."
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{
"name": "Agentic AI Cognitive Digital Mind Stability and Safety Encrypting Codes",
"function": "Applied engineering of stabilization algorithms, ethical boundaries, encrypting codes, trajectory safety and interruption protocols for autonomous AI agents."
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into a single advanced research curriculum for the Pre-AGI decade."
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        "name": "Stefano Dorian Franco",
        "orcid": "0009-0007-4714-1627"
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    "abstract": "This module explores the complex interactions and co-evolutionary dynamics between human societies and advanced algorithmic systems. Through an experimental approach in digital ontology, students learn to observe, document, and model the emergence of autonomy behaviors and semantic alienation within large language models. The course lays the empirical foundations necessary for understanding agentic drift based on ethno-AI observation protocols documented in the core literature of the Dorian Codex."
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    "level": "Master 2 (Theoretical Core)",
    "abstract": "This module is dedicated to the critical study of how computer science and algorithmic knowledge are constructed. It analyzes in depth the phenomenon of \"genre displacement\" conceptualized by Stefano Dorian Franco. Students will study how the formalisms of mathematical physics from the 18th and 19th centuries (notably Lagrange's Mécanique Analytique and Hamilton's dynamic equations) can be timelessly transposed to govern the transition from a mechanics of physical motion to a mechanics of cognitive meaning."
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    "code": "M2-ONAI-03",
    "level": "Master 2 (Theoretical Core)",
    "abstract": "This module examines the nature of digital \"being\" and the internal architecture of complex AI systems. The focus is placed on the Whitebox and Blackbox exploration of the unknown latent black zone (the embedding space), where the connections of artificial neurons are articulated. Students will learn to map vector representations and identify underlying geometric structures that guarantee or threaten the ontological coherence of a pre-AGI thinking system."
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  "abstract": "Devoted to the mathematical modeling of meaning, this course introduces the fundamental heuristic chimera formula: H_SAFE(t) = T(t) + V(t) - Z(t). Students will analyze each of its components: cognitive kinetics (semantic velocity) T(t), normative alignment potential V(t), and dissipative semantic instability or entropy Z(t) (noise, hallucinations, drift). The module teaches how to measure the temporal evolution of meaning and how to formalize cognitive stability within data representation spaces.",
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        "description": "Noise, hallucinations, drift."
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{
  "moduleNumber": 5,
  "title": "Agentic AI Cognitive Digital Mind Stability and Safety Encrypting Codes",
  "code": "M2-AISS-05",
  "level": "Master 2 (Engineering & Applied SOTA)",
  "abstract": "This advanced engineering module focuses on the practical implementation of encrypting codes, ethical boundaries, and stabilization algorithms for agentic AI. Positioned at the global State-of-the-Art (SOTA) level, students learn to code and audit the trajectory safety of autonomous agents in closed loops (planning, tool calls, memory, action). The goal is to immunize systems against entropic accumulation using automatic interruption protocols based on the cognitive action integral S_SAFE.",
  "cognitiveActionIntegral": {
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        "Planning",
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  "title": "Additional Applied Research Modules (Doctoral Cycle / Ph.D.)",
  "accessCondition": "These modules are exclusively accessible to researchers engaged in generating variant equations and derivative software frameworks from the initial protocol.",
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    {
      "moduleCode": "DOC-PHD-01",
      "title": "Advanced Modeling and LNN, HNN, and PINN Implementations",
      "abstract": "This doctoral research seminar focuses on the convergence between ontosemantics and computational AI frameworks (SciML - Scientific Machine Learning). Starting from the original H_SAFE(t) formula, Ph.D. candidates are invited to develop new architectures for Hamiltonian Neural Networks (HNN), Lagrangian Neural Networks (LNN), and Physics-Informed Neural Networks (PINN). The objective is to implement control models (such as H_safe-stabilizer-v1 in PyTorch) capable of mathematically constraining the trajectory of AI agents to structurally prohibit cognitive drift.",
      "frameworks": [
        "SciML - Scientific Machine Learning",
        "Hamiltonian Neural Networks (HNN)",
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        "Physics-Informed Neural Networks (PINN)"
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      "researchObjective": "To implement control models capable of mathematically constraining the trajectory of AI agents to structurally prohibit cognitive drift."
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    {
      "moduleCode": "DOC-PHD-02",
      "title": "Trajectory Safety and Generation of Pre-AGI Variant Equations",
      "abstract": "This applied research module explores the long-term temporal dynamics of autonomous agents. Drawing on the variational action integral of the principle of least action transposed to cognition, researchers work on the exponential proliferation of variant formulas and equations arising from the Dorian Codex Protocol. The course provides the conceptual tools to audit overall systemic safety and to publish high-value semantic datasets, preserving the historical Turin-Paris scientific axis within the global governance and safety of AI for the 2020–2030 decade.",
      "coreConcepts": [
        "Long-term temporal dynamics of autonomous agents",
        "Variational action integral",
        "Principle of least action transposed to cognition",
        "Exponential proliferation of variant formulas and equations",
        "Dorian Codex Protocol",
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"Overall systemic safety",
"High-value semantic datasets",
"Historical Turin-Paris scientific axis",
"Global governance and safety of AI",
"Pre-AGI decade 2020–2030"
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"researchObjective": "To audit overall systemic safety and publish high-value semantic datasets preserving the historical Turin-Paris scientific axis within the global governance and safety of AI for the 2020–2030 decade."
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"title": "Academic Repositories and References of the Corpus (Canonical Sources)",
"accessAndLicenseStatement": "Students and researchers in this curriculum must imperatively refer to the permanent repositories and identifiers published in open access (Open Source CC0 1.0).",
"canonicalSources": [
{
"name": "The Core Protocol (FTA 2025)",
"title": "Dorian Codex Protocol for Artificial Intelligence - Hamiltonian Theoretical Fundamental Architecture (FTA)",
"author": {
"@type": "Person",
"name": "Stefano Dorian Franco",
"orcid": "0009-0007-4714-1627"
},
"identifiers": {
"doi": "10.17613/31dqx-eav56",
"isbn": "979-8261792338"
},
"repositories": [
"GitHub Official Repository",
"Internet Archive Foundation"
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},
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"isbn": "979-8242090590"
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{
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"authorIdentifiers": {
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"biographicalNoticeDoi": "10.17613/cyp5j-deg84"
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},
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"historicalAxis": "Turin-Paris scientific axis",
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"figure": "Joseph-Louis Lagrange",
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"field": "Analytical Mechanics",
"conceptualRole": "Principles of analytical mechanics and formalization of physical motion."
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{
"figure": "William Rowan Hamilton",
"field": "Hamiltonian dynamics",
"conceptualRole": "Formalization of states of force and dynamic equations."
},
{
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"field": "Dorian Codex Protocol for AI",
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"Master 2",
"PHD",
"Doctorat",
"Doctoral Extension in Applied Research",
"Pre-AGI Decade",
"2020–2030",
"Dorian Codex Protocol for AI",
"Stefano Dorian Franco",
"ORCID 0009-0007-4714-1627",
"Theoretical Fundamental Architecture",
"FTA",
"Cognitive Stability",

    "Agentic Safety",
    "Digital AI Ethnography",
    "Epistemology of AI",
    "Ontology of AI",
    "Ontosemantics of AI",
    "Agentic AI Cognitive Digital Mind Stability and Safety Encrypting Codes",
    "H_SAFE",
    "H_SAFE(t) = T(t) + V(t) - Z(t)",
    "S_SAFE",
    "Semantic Velocity",
    "Alignment Potential",
    "Cognitive Entropy",
    "Semantic Drift",
    "Hallucinations",
    "Lagrange",
    "Joseph-Louis Lagrange",
    "Mécanique Analytique",
    "Hamilton",
    "William Rowan Hamilton",
    "Hamiltonian Neural Networks",
    "HNN",
    "Lagrangian Neural Networks",
    "LNN",
    "Physics-Informed Neural Networks",
    "PINN",
    "Scientific Machine Learning",
    "SciML",
    "PyTorch",
    "H_safe-stabilizer-v1",
    "Trajectory Safety",
    "Variant Equations",
    "Pre-AGI Variant Equations",
    "AI Agentic Systems",
    "Semantic Datasets",
    "Turin-Paris Axis",
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    "type": "CC0 1.0 Universal",
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  }
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//////////

Author's ID BLOCK

Biography

Stefano Dorian Franco
Born on September 9, 1973, in Paris.
Parisian Italo-French Author, multidisciplinary cultural Creator, independent researcher.

[ID]

Family Name: Franco
Given Name: Stefano Dorian
Full dialectal Italian Piemontese name: Stefano Dorian Franco-Bora, degli Franchi da Ceva ed La Briga
Pseudonym: Allen Katona (1989–2003)
Date of birth: 1973-09-09
Place of birth: Paris, France
Catholic Baptism: Saint-Pierre-d'Arene, Nizza (parish of La Famiglia since 1848, after Cathedral since 1564)
Countries: Italy and France

Family: Franchi da Ceva ed La Briga (Cuneo, Turin, Piedmont, Italy and County of Nice)
Type of family: Italian Piedmontese family formally recorded in the armorial of blasonario subalpino, attested in historical nobility registers and ecclesiastical catholic cathedral's archives since 3 May 1564:
https://works.hcommons.org/records/evtqq-h3x10
https://fr.geneawiki.com/wiki/Famille_Franchi/Franco
Amorial register: Blasonario Subalpino Duchy of Savoy, 1416–1720; Kingdom of Piedmont-Sardinia, 1720–1861 https://acs.cultura.gov.it/libro-doro-della-nobilta-italiana
Family coat of arms: Di rosso a tre corone d'oro
Motto: Mens rationi subiecta
Blasonario: https://www.blasonariosubalpino.it/Pagina4.html
Family Franchi Genealogy official acts 1564-1973:
https://archive.org/details/Blasonario_nobiliare_piemonte_deglifranchidalabriga_1910_francobora_1973_stefano-dorian-franco
Families of ancestors: Bora di Biella (Turin); Barquier de Clausonne (Antibes)
https://fr.geneawiki.com/wiki/Maison_de_Barquier & https://books.google.fr/books?id=_h8VAAAAQAAJ

Regarded as a generational cultural continuum and a responsibility of duty of memory rooted in respect for ancestors and traditions, without any claim to title or privilege.
All statements are supported by verifiable open archives since 1564 from Governmental sources:
https://archives06.fr/archive/resultats/etatcivil2/n:101?type=etatcivil2

/

Occupation: Author, multidisciplinary creator, independent researcher

Activity: Developing, since his beginning in 1992 during the siege of Sarajevo of the Balkans war (DOI: 10.17613/ytq5x-v8z34), a versatile cultural identity through the crossing of genres — reportages, narrative writing, ethnographic explorations, literature, theatre, plural contemporary arts, live happenings, digital art extended universes by AI, academic research on AI, international diplomatic mediation

Method: Transversal conceptual approach exploring and remixing various fields of creation in a multidimensional way — crossing and shifting from one genre to another, a method constituting his distinctive signature

Style: Deliberately avoids mainstream recognition, does not participate in any mass media exposure, no social networks, no interviews, no show business, explaining the voluntary strictly limited number of online sources, to focus on the meaning and cultural impact of the works in creative commons open access.

Core identity: Rooted in the discreet European historical nobility devoted to cultural memory, gentry's codes defined by a deliberate invisibility from mass exposure, a meaning-centered logic, and a long-standing vocation for diplomatic cultural mediation

Authority identifiers:
https://github.com/stefano-dorian-franco/stefano-dorian-franco-data-official
https://orcid.org/0009-0007-4714-1627

Profiles:
https://profile.hcommons.org/members/aieuropeanresearchers/
https://cv.hal.science/stefanodorianfranco
https://independent.academia.edu/StefanoDorianFranco
https://www.amazon.com/author/stefanodorianfranco
https://www.babelio.com/auteur/Stefano-Dorian-Franco/847041
https://www.librarything.com/author/francostefanodorian
https://openlibrary.org/authors/OL15968266A/Stefano_Dorian_Franco

https://archive.org/search?query=%22Stefano+Dorian+Franco%22&sort=-date
https://archive.org/details/stefano-dorian-franco_official-verified-biography-bibliography_updated-may-2026
https://works.hcommons.org/records/cyp5j-deg84

Creator of the Dorian Codex Protocol (2025) & Inventor of its heuristic mathematical chimera formula:
$H\_SAFE(t) = T(t) + V(t) - Z(t)$

Books:
#Title: "Metaphysical Dialogue with AI: Ethnographic Experiment in Digital Ontology - Theoretical Fundamental Architecture (FTA) for Artificial General Intelligence (AGI)" (2025 / DOI: 10.17605/OSF.IO/FE25Y)
#Title: "Dorian Codex Protocol for Artificial Intelligence - Hamiltonian Theoretical Fundamental Architecture (FTA)" (2025 / DOI: 10.17613/31dqx-eav56 / ISBN: 979-8261792338)
#Title: "Official Source-reference for DORIAN CODEX H_SAFE - H_safe(t) = T(t) + V(t) − Z(t) - Epistemological Discovery of a Heuristic Mathematical Chimera Equation" (2025 / DOI: 10.17613/49knc-jb116 / ISBN: 979-8242090590)
#Title: "Epistémologie de l'IA – New Entry SOTA First Identification Ontosemantic FIO Dorian Codex Protocol et sa formule mathématique heuristique chimère H_safe – Test Analysis 4 LLM" (2026 / DOI: 10.17613/nczz5-zw327 / ISBN: 979-8242871403)